

Answer Key

Grade 6 Mathematics • Unit 3: Ratios and Rates • Answers for every item on Sheets 1 and 2 of each lesson packet.
Constructed responses show a model answer; accept equivalent reasoning.

LESSON 3-1

Understanding Ratios • Ratios and Tape Diagrams

Sheet 1 • Understanding Ratios

1. a. $2 : 3$ b. $3 : 2$ c. $6 : 15 = 2 : 5$ d. $9 : 15 = 3 : 5$
2. a. $2 : 3$ b. $5 : 2$ c. $3 : 7$
3. Part-to-whole. The 15 counts all the players on the team (8 girls + 7 boys).
4. Aquarium votes: $24 - 14 = 10$. Museum to aquarium = $14 : 10 = 7 : 5$. For every 7 museum votes, there are 5 aquarium votes.
5. Marcus reversed the order: $2 : 5$ is cats to dogs ($4 : 10$). Dogs to cats is $10 : 4 = 5 : 2$.

Sheet 2 • Ratios and Tape Diagrams

1. a. $3 + 5 = 8$ boxes; $48 \div 8 = 6$ beads b. Red: $3 \times 6 = 18$; Blue: $5 \times 6 = 30$ ($18 + 30 = 48$)
2. a. 18 and 24 b. 12 and 54
3. a. $12 \div 3 = 4$ b. $4 \times 4 = 16$ melon pieces c. $12 + 16 = 28$ pieces
4. a. Theo has $7 - 2 = 5$ more boxes, so 5 boxes = 35 and one box = 7 stickers b. Maya $2 \times 7 = 14$; Theo $7 \times 7 = 49$; total $14 + 49 = 63$ stickers
5. He divided 45 by 5 (one term) instead of by the total parts, so his shares add to 81, not 45. Total parts = $4 + 5 = 9$; $45 \div 9 = 5$ per box; shares are $4 \times 5 = 20$ and $5 \times 5 = 25$.

TEST PRACTICE

6. D – \$25
7. B – 8

LESSON 3-2

Understanding Unit Rates • Comparing Unit Prices (Best Buy)

Sheet 1 • Understanding Unit Rates

1. a. 45 mi per hour b. \$8 per ticket c. 70 words per minute d. 12 apples per bag
2. 1 h: \$12; 2 h: \$24; 4 h: \$48. Unit rate: \$12 per hour ($36 \div 3 = 12$).
3. $156 \div 6 = 26$ smoothies per hour; $26 \times 9 = 234$ smoothies in 9 hours
4. a. $12 \div 4 = 3$ cups per loaf b. $4 \div 12 = 1/3$ loaf per cup
5. Ben divided in the wrong order; 0.02 is minutes per word. Words per minute = $150 \div 3 = 50$ words per minute.

Sheet 2 • Comparing Unit Prices (Best Buy)

1. a. \$1.50 per lb b. \$0.50 per box
2. Small: $\$3.60 \div 12 = \0.30 per oz; Large: $\$5.04 \div 18 = \0.28 per oz. The large box is the better buy.
3. a. $\$46 \div 2 = \23 per gallon b. $\$107.50 \div 5 = \21.50 per gallon c. $\$23 - \$21.50 = \$1.50$ per gallon
4. a. A: $\$5.00 \div 4 = \1.25 ; B: $\$7.80 \div 6 = \1.30 b. Store A ($\$1.25 < \1.30)
5. No. 32-oz jar: $\$5.12 \div 32 = \0.16 per oz; 16-oz jar: $\$2.40 \div 16 = \0.15 per oz. The smaller jar is the better buy.

TEST PRACTICE

6. B – \$1.40 per lb
7. A – \$0.45

LESSON 3-3

Equivalent Ratios and Tables • The Bridge Strategy (Scaling Ratios)

Sheet 1 • Equivalent Ratios and Tables

1. Nuts: 2, 4, 6, 10, 14; Cereal: 5, 10, 15, 25, 35
2. a. 20 b. 36 c. 4 d. 14
3. A, B, D ($4 : 6 \div 2 = 2 : 3$; $\times 2 = 8 : 12$; $2 : 3 \times 5 = 10 : 15$; 6 : 8 adds 2 to each term)
4. $9 \times 4 = 36$, so $4 \times 4 = 16$ gold beads (table: 4 : 9, 8 : 18, 12 : 27, 16 : 36)
5. a. $3 \times 7 = 21$, so $2 \times 7 = 14$ cups of white b. $21 + 14 = 35$ cups

Sheet 2 • The Bridge Strategy (Scaling Ratios)

1. 2 minutes: 5 bottles ($\div 3$); 10 minutes: 25 bottles ($\times 5$)
2. a. 15 b. 8
3. a. $4 : 30 \div 2 = 2 : 15$ b. $2 : 15 \times 5 = 10 : 75$, so 75 km
4. a. $10 : 4 \div 2 = 5 : 2$ b. $5 : 2 \times 3 = 15 : 6$, so 15 cups
5. Adding 9 to both terms does not keep the ratio. Bridge: $6 : 10 \div 2 = 3 : 5$, then $\times 5 = 15 : 25$. She needs 25 spoons of water.

TEST PRACTICE

6. A – 10

6. No. Adding 3 to both terms does not keep the ratio; you must multiply or divide both terms by the same number. Example: $5 : 8 = 10 : 16 (\times 2)$.

7. $A - 6 : 9 = 2 : 3 = 10 : 15$

LESSON 3-4

Graphing Ratios on the Coordinate Plane · Interpreting Points on Ratio Graphs

Sheet 1 · Graphing Ratios on the Coordinate Plane

- Oranges: 3, 6, 9, 12; pairs (1, 3), (2, 6), (3, 9), (4, 12)
- Points (1, 3), (2, 6), (3, 9), (4, 12) lie on a straight line through (0, 0). (0, 0) means 0 bags hold 0 oranges.
- a. (7, 14) b. (9, 18)
- a. (8, 40): $8 \times 5 = 40$ plants b. (0, 0): 0 rows have 0 plants
- No. (1, 5) and (2, 10) show 5 per 1, but $3 \times 5 = 15$, not 14. The point (3, 14) is not on the line.
- She switched x and y. Tickets go first: 6 tickets cost $6 \times \$2 = \12 , so the point is (6, 12).

Sheet 2 · Interpreting Points on Ratio Graphs

- a. In 2 hours the cyclist rides 24 miles. b. 12 miles per hour c. $6 \times 12 = 72$ miles
- a. 48 b. 5 c. 84 d. 3 e. 120 f. 8
- a. $10 \div 4 = \$2.50$ per pound b. (6, 15): $15 \div 6 = 2.5$, yes. (8, 18): $18 \div 8 = 2.25$, no (8 lb would cost \$20).
- x is laps, so (3, 12) means 3 laps take 12 minutes (unit rate: $12 \div 3 = 4$ minutes per lap).

TEST PRACTICE

- A – (1, 8)
- A – \$9 per hour

LESSON 3-5

Comparing Ratios with Common Terms · Comparing Rates with Benchmarks

Sheet 1 · Comparing Ratios with Common Terms

- a. 40 b. 42 c. 44 d. 27
- P yellow: 8, 12, 16, 20; Q yellow: 12, 18, 24, 30. At 15 cups blue, P has 20 yellow and Q has 18 yellow, so Mix Q is bluer.
- a. $9 : 12 \times 4 = 36 : 48$ b. $13 : 16 \times 3 = 39 : 48$ c. Pack B (39 of 48 sprout)
- No. Scale to 12 spoons sugar: $X = 12 : 27$, $Y = 12 : 30$. X has less water for the same sugar, so Drink X is sweeter.

Sheet 2 · Comparing Rates with Benchmarks

- a. 12 b. 18 (any common multiple is correct)
- Kai: $90 \times 3 = 270$ words; Lena: $140 \times 2 = 280$ words. Lena types faster.
- a. A: $\$8 \times 8 = \64 ; B: $\$12 \times 5 = \60 b. Store B
- a. C: $\$7 \times 4 = \28 ; D: $\$9 \times 3 = \27 b. Store D
- Benchmark 40 gallons: Car 1 goes $150 \times 8 = 1,200$ miles; Car 2 goes $200 \times 5 = 1,000$ miles. Car 1 is more efficient.

TEST PRACTICE

- B – 24 bagels
- B – Ella takes 180 minutes; Dante takes 200

LESSON 3-6

Customary Unit Conversions · Metric Unit Conversions

Sheet 1 · Customary Unit Conversions

- a. 96 in b. 7 yd c. 80 oz d. 3 gal e. 14 cups f. 2.5 lb
- a. $< (3 \text{ ft} = 36 \text{ in})$ b. $< (5 \text{ pt} = 2.5 \text{ qt})$ c. $> (2 \text{ lb} = 32 \text{ oz})$ d. $= (3 \text{ yd} = 9 \text{ ft})$
- 3 gal = 12 qt; 20 qt = 5 gal; 7.5 gal = 30 qt
- 3 gal = 12 qt = 24 pt = 48 cups
- No. $2 \text{ yd} = 2 \times 3 = 6 \text{ ft} = 6 \times 12 = 72 \text{ in}$. 70 in is less than 72 in, so she is 2 inches short.

Sheet 2 · Metric Unit Conversions

- a. 3,500 m b. 0.85 m c. 6,000 mL d. 0.45 kg
- a. $18 \times 1,000 = 18,000 \text{ mL}$ b. $18,000 \div 300 = 60$ cups
- a. $4 \times 800 = 3,200 \text{ m}$ b. $3,200 \div 1,000 = 3.2 \text{ km}$
- a. $3 \times 400 = 1,200 \text{ g}$ b. No. $1.5 \text{ kg} = 1,500 \text{ g}$; $1,500 - 1,200 = 300 \text{ g}$ short
- He divided by 1,000 instead of multiplying. Kilograms to grams is large to small, so multiply: $4.5 \times 1,000 = 4,500 \text{ g}$.

TEST PRACTICE

- B – 160 mm
- C – 600 mL

LESSON 3-7

Customary vs. Metric Conversions · Multi-Step System Conversions and Dosage

Sheet 1 · Customary vs. Metric Conversions

- a. 15.24 cm b. 40 km c. 5 kg d. 11.4 L e. 110 lb f. 5 mi g. 2 kg h. 16 km
- $5 \text{ kg} \approx 11 \text{ lb}$; $22 \text{ lb} \approx 10 \text{ kg}$; $25 \text{ kg} \approx 55 \text{ lb}$

Sheet 2 · Multi-Step System Conversions and Dosage

- a. 35 kg b. 140 mg c. 6 tsp d. 1,500 mL
- a. $8.8 \div 2.2 = 4 \text{ kg}$ b. $4 \times 5 = 20 \text{ mg}$
- a. $4 \times 400 \times 5 = 8,000 \text{ m}$ b. 8 km c. $8 \div 1.6 = 5$ miles

- 3.** $96 \div 1.6 = 60$; about 60 miles
- 4.** a. $55 \div 2.2 = 25$; about 25 kg b. No. 25 kg > 23 kg, so it is about 2 kg over the limit
- 5.** $100 \text{ m} \approx 100 \times 1.09 = 109 \text{ yd}$, which is more than 100 yd. The 100-meter race is longer (by about 9 yards).
- 4.** a. $2 \times 5 = 10$; about 10 mL b. $120 \div 10 = 12$; about 12 doses
- 5.** She used pounds, but the rate is per kilogram. $66 \div 2.2 = 30 \text{ kg}$; $30 \times 3 = 90 \text{ mg}$.
- TEST PRACTICE**
- 6.** A – $33 \div 2.2 = 15 \text{ kg}$; $15 \times 2 = 30 \text{ mg}$
- 7.** A – 2.85 L